

AI4TELCO HACKATHON · MTN NIGERIA CHALLENGE

RiskGuard AI

The agentic AI risk officer that watches every network, billing, and customer signal — and acts before the outage.

Detects

Predicts

Mitigates

Reports to NCC

Built on Microsoft Azure OpenAI + Semantic Kernel · AI Skills Week Lagos 2026

Microsoft · MTN Foundation Nigeria · Data Science Nigeria

Risk hides between the silos.

"Telecom operators rely on multiple, siloed data sources across network, sales, devices, and customer channels — making it difficult to identify operational risks early."

— MTN Nigeria, AI4Telco problem statement



Network & BTS

Cell health, throughput, latency



Billing & Revenue

Recharge, top-ups, leakage flags



Customer Channels

Complaints, social, contact centre



Devices & Sales

Activations, SIM swaps, IMEI

Result: service disruption · revenue leakage · customer churn · NCC penalties

Operators learn from Twitter — not dashboards.

47 min

average operator delay between first signal and confirmed incident

NGN-bn

regulatory penalties NCC has issued operators for QoS breaches in recent years

1 in 3

Nigerian subscribers say poor service is their #1 reason to switch network

Today's tools are dashboards. Dashboards wait to be looked at. Risk doesn't.

RiskGuard AI — an AI risk officer that takes action.

Not a dashboard. An always-on agent that **reads** every signal across network, billing, sales and customer channels, **predicts** the next breach, **recommends** a fix, and **files the NCC report** — with a human always in the loop.

1



Sense

Ingest 7 signal domains in real time

2



Reason

Agentic LLMs reason over grounded tools

3



Recommend

Compare actions vs. do-nothing, ranked

4



Report

Auto-assemble NCC-ready evidence pack

Four layers. One Microsoft AI stack.



1. Signal Ingest

Network, BTS, billing, recharge, sales, complaints, devices — normalized into one stream.



2. Risk Engine

Rolling z-scores, anomaly + co-occurrence, per-LGA score, time-to-breach, impact model.



3. Agentic Copilot

Semantic Kernel + Azure OpenAI. 5 role agents call grounded tools. ClaimValidator gates output.



4. Operator Console

Live risk map · incident panel · mitigation simulator · approval flow · NCC compliance pack.

From green to recovered in 90 seconds.



00:00

Baseline

All LGAs green



00:15

Trigger

Ikeja BTS deviation +
complaint spike



00:30

Risk = 87

47 min to SLA breach ·
18,420 subs



01:00

Operator approves

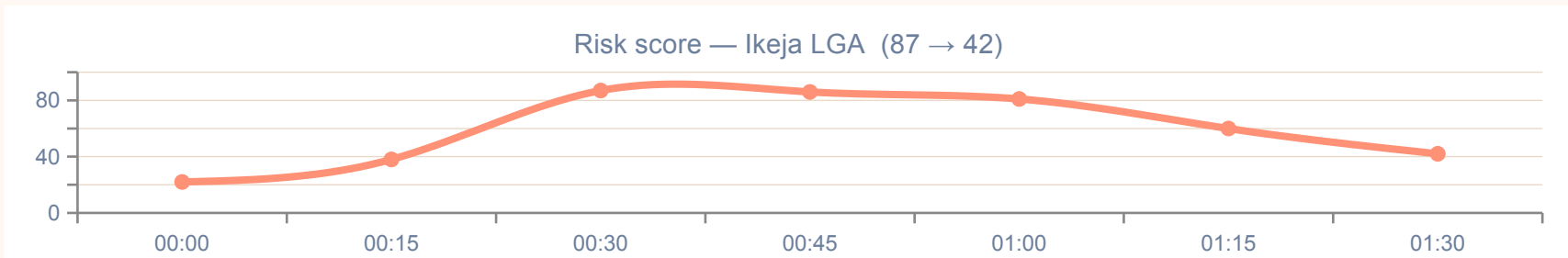
Reroute traffic · audit
logged



01:30

Risk = 42

Recovery confirmed · NCC
pack ready



Five specialist agents. Zero hallucinations.



Network Agent

Localizes the fault, pulls BTS evidence



Revenue Agent

Spots leakage in recharge & billing streams



Customer Agent

Reads complaints, projects churn risk



Mitigation Agent

Simulates options, ranks by impact + cost



Compliance Agent

Drafts NCC notes, assembles evidence pack

Every claim — every KPI, naira figure, subscriber count — must trace to a tool call. ClaimValidator strips the rest.

Five agents. Pluggable tools. Any data system.

AGENT	TOOL (SK function)	EXTERNAL SYSTEM
 Network Agent	▪ <code>get_incident_context()</code> ▪ <code>get_signal_evidence(net, bts)</code>	Network OSS · NMS · BTS <i>Ericsson · Huawei · Nokia</i>
 Revenue Agent	▪ <code>get_signal_evidence(billing, recharge)</code> ▪ <code>estimate_impact()</code>	Billing · Charging · Top-up <i>Amdocs · BSCS · Comviva</i>
 Customer Agent	▪ <code>get_signal_evidence(complaints, social)</code> ▪ <code>estimate_impact()</code>	CRM · Contact Centre · Social <i>Salesforce · Genesys · X (API)</i>
 Mitigation Agent	▪ <code>get_mitigation_playbook()</code> ▪ <code>run_pre_action_simulation()</code>	Playbook DB · Risk Engine <i>Custom · ServiceNow</i>
 Compliance Agent	▪ <code>get_audit_trail()</code> ▪ <code>generate_ncc_pack_draft()</code>	Audit Log · NCC e-filing <i>Internal · NCC Portal</i>

Adapter pattern: tools are thin connectors. Swap MTN systems for Airtel, Glo, or any operator — no agent code changes.

Trustworthy enough for regulators and CEOs.



Grounded answers

Agents only speak from tool results.
No CSV browsing. No free-form claims.



Human-in-the-loop

No mitigation runs without operator approval.
The approval IS the trigger.



Append-only audit

Every action is logged: who, when, why,
expected impact, actual outcome.

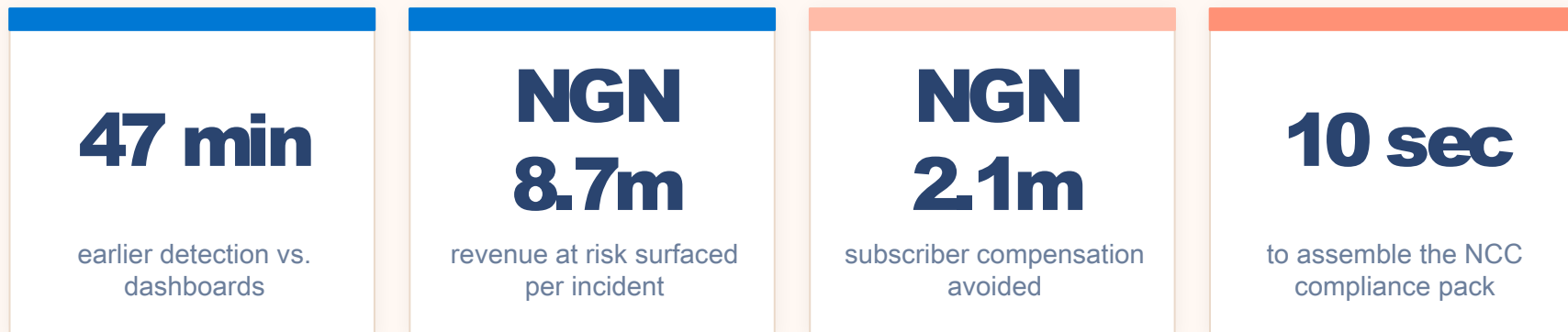


Regulator-ready output

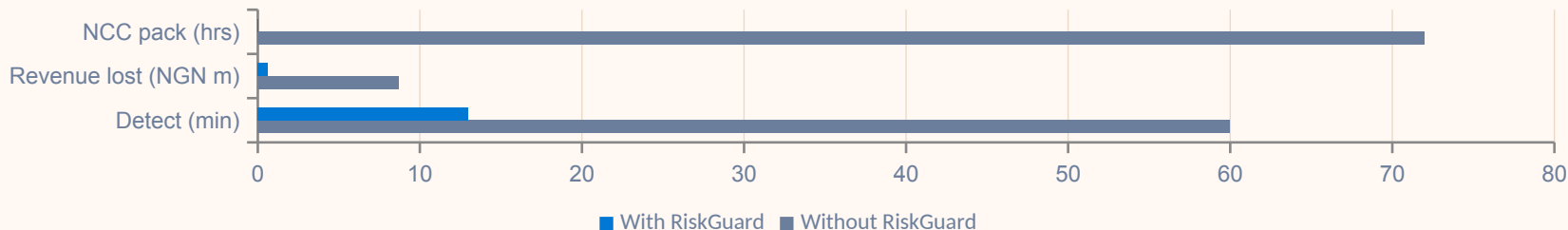
NCC-format pack assembles in seconds,
straight from the audit trail.

Autonomous enough to be useful. Constrained enough to be deployed.

What one Ikeja-class incident looks like.



Per-incident operator outcome



Not a chatbot. Not a BI tool. A new category.

Legacy NMS Network monitoring	Generic LLM chat ChatGPT-style	RiskGuard AI Agentic risk officer
✗ Single-domain visibility ✗ Static thresholds ✗ No mitigation guidance ✗ Manual NCC reporting	✗ Hallucinates KPIs ✗ No live data tools ✗ No audit trail ✗ Not regulator-safe	✓ 7 signal domains fused ✓ Predicts breach 47 min ahead ✓ Recommends + simulates fix ✓ Auto-files NCC pack

And we built it on the host's stack: Microsoft Azure OpenAI + Semantic Kernel.

From hackathon to West African telco standard.

Now

Vertical slice live

Ikeja demo. Real risk engine, agentic copilot, NCC pack. End-to-end.

90 days

Pilot with 1 MNO

Connect to one operator's read-only data lake. One Lagos region. Measure detection lift.

12 months

Multi-LGA SaaS

Cover all 20 Lagos LGAs. Add fraud + SIM-swap agents. SOC2-track.

24 months

Regional platform

Tier-2/3 operators across West Africa. Per-subscriber pricing. NCC + ARCEP-ready.

Business model: SaaS license per operator + per-incident NCC compliance fee. Avoids one ₦ 100m fine = pays for the year.

LET'S BUILD TELCO RESILIENCE FOR NIGERIA

RiskGuard AI

Detect risk earlier. Mitigate it faster. Prove it to the regulator.

THE TEAM

Ifunanya

Team Lead · Product & Strategy

Steers the mission. Owns the pitch. Sets the calibre.

Favour

Data & Risk Engine

Ladipo

API, State, Compliance

Marvelous

Frontend Dashboard

Tobi

Agentic Copilot

Our ask: the win, a pilot conversation with MTN, and the runway to ship this to production.